

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 7: Ratio and Proportion

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Properties of proportion

1. Use the properties of proportion to solve $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$, where $x \neq 0$.

ICSE 2025, 3 marks

SOLUTION

Apply dividendo: if $\frac{a}{b} = \frac{c}{d}$ then $\frac{a-b}{b} = \frac{c-d}{d}$.

$$\frac{6x^2 + 3x - 5 - (3x - 5)}{3x - 5} = \frac{9x^2 + 2x + 5 - (2x + 5)}{2x + 5}, \text{ so } \frac{6x^2}{3x - 5} = \frac{9x^2}{2x + 5}$$

Since $x \neq 0$, divide both sides by $3x^2$: $\frac{2}{3x - 5} = \frac{3}{2x + 5}$

$$4x + 10 = 9x - 15, \text{ so } 5x = 25.$$

Answer: $x = 5$

2. Given $\frac{(a+b)^3}{(a-b)^3} = \frac{64}{27}$, find (a) $\frac{a+b}{a-b}$ (b) the ratio $a : b$, using the properties of proportion.

ICSE 2024, 4 marks

SOLUTION

(a) Take the cube root of both sides: $\frac{a+b}{a-b} = \frac{4}{3}$

(b) Apply componendo and dividendo: $\frac{(a+b) + (a-b)}{(a+b) - (a-b)} = \frac{4+3}{4-3}$

$$\frac{2a}{2b} = \frac{7}{1}$$

Answer: (a) $\frac{4}{3}$ (b) $a : b = 7 : 1$

2 Proportion

3. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in x s and y m in 5 s. Find x and y . (a) $x = 4, y = 150$ (b) $x = 3, y = 100$ (c) $x = 4, y = 100$ (d) $x = 3, y = 150$ ICSE 2024, 1 mark

SOLUTION

Speed = $\frac{60}{2} = 30$ m/s. At uniform speed, distance is proportional to time.

$$x = \frac{90}{30} = 3 \text{ and } y = 30 \times 5 = 150$$

Answer: (d)

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